MD. Sazzad Hossain Adib

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EDUCATION

North South University *Sep 2021 – Dec 2025*
Bachelor of Science in Computer Science and Engineering *Dhaka, Bangladesh*
CGPA: 3.83/4.00 | Merit Scholarship (Top 1% of class)

PROJECTS

AI Doctor Assistant (SaaS Web Application) | *Next.js, TypeScript, NEON PostgreSQL, Clerk, Drizzle ORM, VAPI, OpenRouter*

- Built a **voice-powered medical consultation platform** providing real-time, AI-driven healthcare assistance
- Implemented **LLM-based symptom analysis** to suggest suitable doctors and enable **natural doctor-patient conversations**
- Integrated **VAPI Voice Assistant** with OpenRouter AI for real-time speech understanding and interaction
- Designed automatic **medical report generation** including patient details, symptoms, prescriptions, advice, and session summary
- Secured platform with **Clerk authentication** and managed scalable data operations using **NEON PostgreSQL + Drizzle ORM**
- Structured with SaaS-ready architecture, featuring dashboards, doctor listings, pricing, health tips, and contact pages
- Deployed on **Vercel** for seamless performance, modern UI, and scalable cloud readiness
- Project GitHub: github.com/sazzadadib/AI-Doctor-Assistant Live Demo: aidocassistant.vercel.app

SHOBDOTORI: Regional-to-Standard Bangla Speech Recognition | *Python, PyTorch, Whisper, Wav2Vec2, KenLM, ASR*

- Developed an **Automatic Speech Recognition (ASR)** system to transcribe **regional Bangladeshi dialects into standard Bangla**
- Built and curated a **3,800+ audio sample dataset** covering **20 regional dialects** across Bangladesh
- Implemented **transformer-based speech models** (Whisper, Wav2Vec2) with **phoneme alignment** and **audio data augmentation**
- Applied **n-gram KenLM post-processing** to improve transcription fluency and linguistic correctness
- Evaluated model performance using **Normalized Levenshtein Similarity (NLS)** and achieved **competitive leaderboard results**
- Developed as an **AI Hackathon project under Televerse 1.0**, organized by **CUET (Department of ETE)**

DeshiPlate AI: Bangladeshi Food Recognition & Nutrition Assistant | *Next.js, TypeScript, NextViT, PyTorch, NEON PostgreSQL, Drizzle ORM, OpenRouter AI*

- Developed an **AI-powered food recognition system** specialized in **33 traditional Bangladeshi dishes**
- Created and curated a **custom dataset of Bangladeshi cuisine** for training the NextViT classification model
- Implemented **NextViT deep learning architecture** achieving **89.76% accuracy** on food classification tasks
- Built real-time **inference engine** enabling users to upload food images and receive instant AI-driven insights
- Integrated **personalized nutrition recommendations and health suggestions** based on classified food items
- Managed scalable data operations using **NEON PostgreSQL + Drizzle ORM**
- Deployed on **Vercel** with modern, responsive UI for seamless user experience
- Project GitHub: github.com/sazzadadib/DeshiPlate-AI Live Demo: deshiplateai.vercel.app

TimeClipAI: Real-Time Action Classification and Time Segmentation | *PyTorch, Anchor Transformers, I3D, Python, Git*

- Developed **TimeClipAI**, a framework for **online temporal action localization** using Anchor Transformers
- Enabled **real-time action classification and time segmentation** in full-length videos
- Integrated support for multiple benchmark datasets including **EGTEA, EPIC-Kitchen 100, THUMOS'14**, and CricShot10
- Leveraged pre-trained **I3D features** for efficient training and testing
- Designed and trained a **post-processing Offset Scoring Network (OSN)** to refine predictions
- Achieved state-of-the-art performance with robust generalization across datasets
- Open-sourced implementation with full pipeline: github.com/sazzadadib/CSE499-TimeClip_AI

Educational Chatbot using RAG | *SLMs, Python, Ollama, LangChain, HuggingFace, FastAPI, Prompt Engineering, Git*

- Developed lightweight SLM-based chatbot for offline, device-efficient use
- Designed user-friendly web interface for better accessibility
- Achieved 75% accuracy on 4,000-question dataset using RAGChecker
- Enabled accessible learning for underserved communities

Road Accident Analysis System | *Python, Scikit-learn, NumPy, Pandas, Git*

- Predicted patient status and injury type using Road Traffic Accident Dataset
- Compared Logistic Regression, Decision Tree, SVM, XGBoost, Random Forest
- Achieved 99.1% accuracy for status, 89% for injury type
- Improved emergency response through data-driven insights

Agile-Driven Movie Reservation Website | *Node.js, React, MongoDB, Postman, Jest, Trello, GitHub Wiki, JSDoc, Git*

- Developed scalable website with 5-member team using Agile Scrum
- Contributed to frontend/backend with MVC architecture
- Documented codebase with JSDoc for maintainability
- Ensured reliability with Jest unit tests

PUBLICATIONS

Z-Pruner: Post-Training Pruning of Large Language Models for Efficiency without Retraining
Samiul Basir Bhuiyan, **Md. Sazzad Hossain Adib**, Mohammed Aman Bhuiyan, Muhammad Rafsan Kabir, Moshiur Farazi, Shafin Rahman, and Nabeel Mohammed
Accepted at IEEE AICCSA 2025
arxiv.org/abs/2508.15828 github.com/sazzadadib/Z-Pruner

EXTRA-CURRICULAR ACTIVITIES & COMPETITIONS

- **National AI Competition Finalist, AI Fication 2025 (Televerse 1.0)**
  Chittagong University of Engineering & Technology (CUET)
  - Focused on regional-to-standard Bangla speech transcription
  - Ranked **4th** in the private online round and **6th overall** in the final stage
- **Top 15 Finalist (out of 100+ teams), NSU Capstone Project Challenge – Innovation Challenge 2025**
  North South University
  - Recognized for developing impact-driven and innovative project solutions

DOMAIN OF INTEREST

Applied AI and Machine Learning:

- Temporal Action Localization in Videos
- Efficient LLM Deployment (Pruning, Quantization)
- Retrieval-Augmented Generation (RAG) Systems
- Small Language Models for Edge AI

Full-Stack & Scalable Systems:

- MERN Stack and API Development
- Agile-Based Scalable Web Applications
- AI-Integrated Web Interfaces
- Cloud-Ready ML Pipelines

SKILLS

Languages & Frameworks: Python, JavaScript, PHP, Node.js, React.js, Next.js, Flask, FastAPI, Express.js
Machine Learning & AI: PyTorch, Scikit-learn, HuggingFace Transformers, LangChain, LlamaIndex, Ollama
AI Techniques: RAG Pipelines, LLM/SLM Fine-tuning, Model Pruning, Prompt Engineering
Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn
Database & DevOps: PostgreSQL, MySQL, MongoDB, REST APIs, Git
Tools & Productivity: GitHub, Trello, JSDoc

REFERENCES

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Dr. Nabeel Mohammed
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